→ chmod umask

→ Modify the permissions of a existing file.

(~ umask ↓ max. perm possible)

↓

permissions given to the newly created file.

→ fcntl :-
→ dup
dup2

} duplicate the file descriptor.

printf → terminal (pts)

printf → ttyS0 (serial port)

open → ttyS0

↳ 3 → 1 →

printf ("Hello world \n");

Duplicating the fd :-

↳ Is to redirect the data flow.

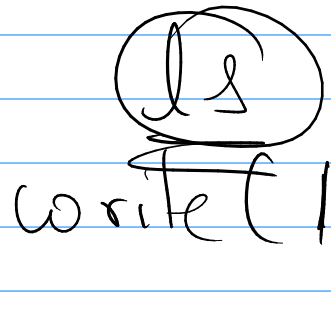
$ ls | wc -l

Shell → 2 process

process 1 process 2

1 → pipe. 0 → pipe-RD

pipe-RD →



$ls

shell → proces

main

↳ 0, 1, 2

file → 3

dup(3) → 4

main( )

{

opened file

↳ 3

close(1)

dup(3) → 1

write(1 → offset → 5 inc

write(3 →

offset changes are going to be applicable for both the descriptor.

main()

↳ 1 → stdout

dup2 (3, 1) ← open 3

↳ open

→ formatting

↳

Slow formatting

fast "

(↳ Slow → filled → 0

fast → file system

→ How many nodes ?

↳ Inode num range

No. of blocks

Each block size

file attrib

file → Inode num → Inode

|

ptr → physical data

→ $vi f1

chmod → Read

Threads

→ Thread is a basic unit of execution.

→ process can be of single threaded & multi threaded.

→ process → multi threaded.

↓

multiple jobs to be done simultaneously.

Zoom

Mic (Audio)

Video

Chat

File Sharing.

Teams.

↓

Mic (Audio)

Video

Chat

File Sharing

→ file Sharing

Mobile → uploaded → cloud

device

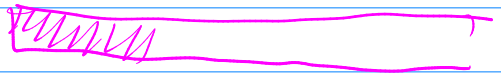
memory

data Tf (100 bytes / 100 bytes)

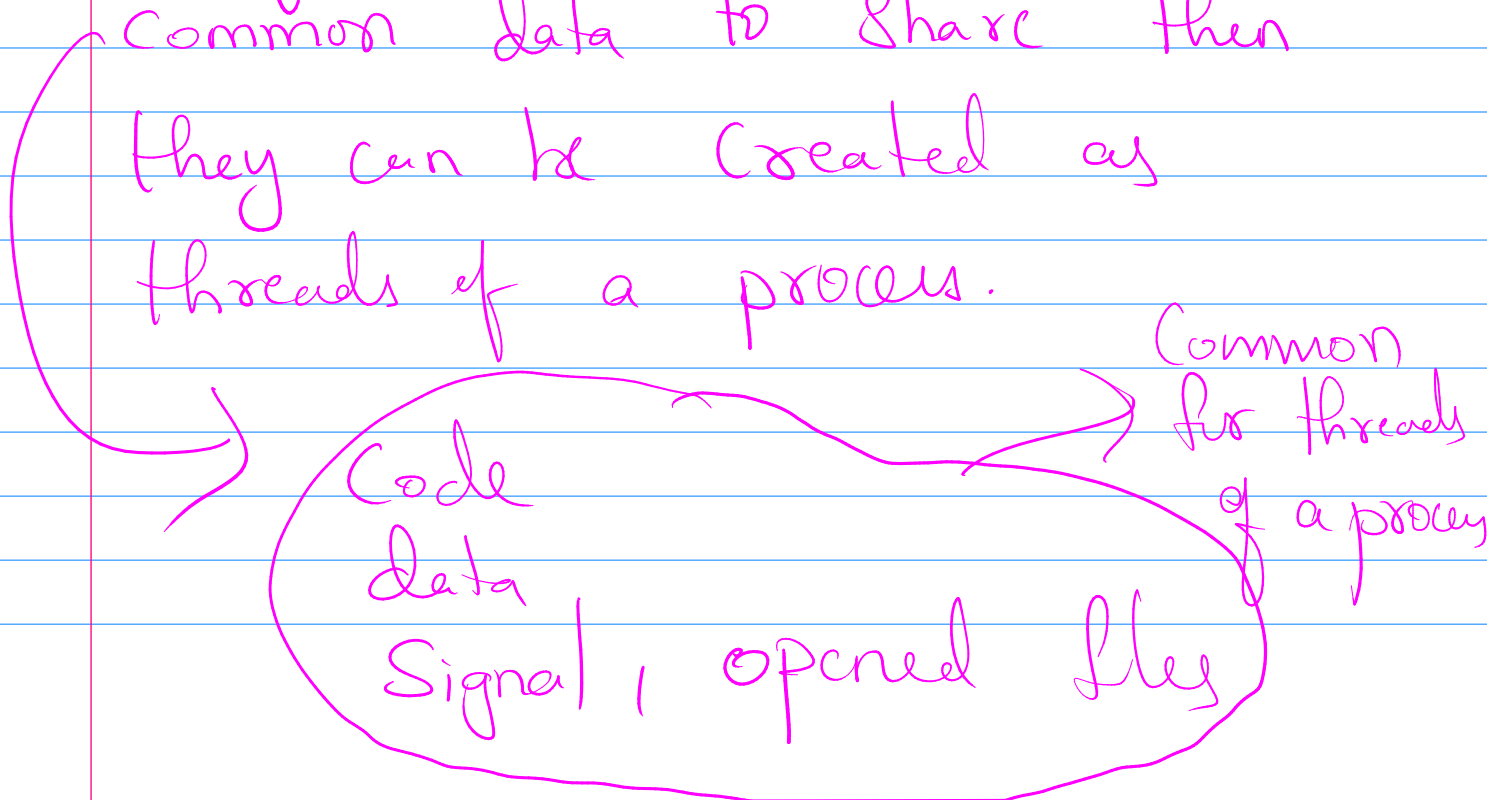
Speed

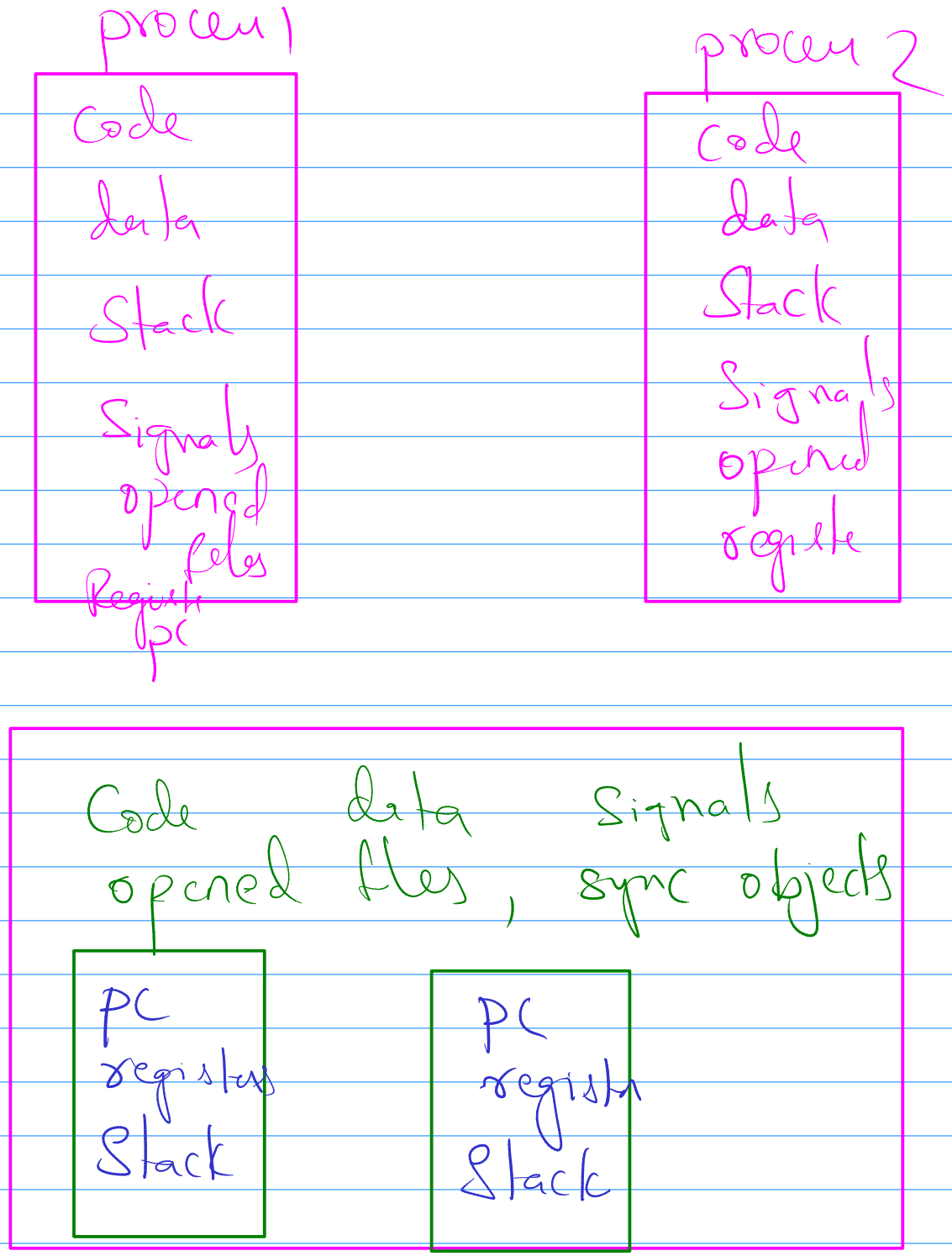
% of data remaining

time remaining



→ If multiple jobs have some common data to share then they can be created as threads of a process.




proc 1
Code
data
Stack
Signals
opened files
Registers
pc
proc 2
Code
data
Stack
Signals
opened
regists
Code
data
Signals
opened files, sync objects
pc
registers
Stack
pc
regists
Stack